

Hemanth Butham

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Data Engineer

Experienced data engineer skilled in designing scalable ETL pipelines and data warehouses using Apache Spark and Medallion Architecture. Proficient in Python, AWS cloud services, and real-time IoT data streaming with strong expertise in machine learning models for fraud detection and predictive analytics. Demonstrated ability to deliver production-ready solutions with automated testing, CI CD, and monitoring. Adept at developing AI benchmarking systems and full stack data analytics platforms.

SKILLS

Programming Languages: Python, SQL, JavaScript, HTML, CSS, Bash, YAML, Dockerfile, JSON.

Data Engineering: ETL, Data Pipelines, Apache Spark, Apache Kafka, Delta Lake, Data Warehousing, Medallion Architecture, Data Quality, Data Analytics.

Machine Learning & AI: Graph Neural Networks, PyTorch Geometric, scikit-learn, Classification Models, Anomaly Detection, Time Series Forecasting, Focal Loss, Model Evaluation, Explainability.

Cloud & DevOps: AWS Cloud, Docker, Docker Compose, Git, GitHub Actions CI/CD, FastAPI, Prometheus, Grafana, Monitoring, Automated Testing.

PROFESSIONAL EXPERIENCE

Data Engineering Intern, EduSkills Foundation® Jul 2024 - Sep 2024 | India

- Strengthened cloud infrastructure knowledge by executing AWS service projects, improving data pipeline efficiency by 20 percent.
- Constructed and optimized ETL workflows, enabling scalable data ingestion and transformation for enhanced processing performance.

Web Full Stack Developer Intern, EduSkills Foundation® Oct 2024 - Dec 2024 | India

- Engineered responsive web interfaces using HTML, CSS, and JavaScript, reducing front-end load times by 15 percent.
- Integrated front-end and back-end components to streamline web application functionality, boosting overall user experience metrics.

AI Agent Evaluation Fellow, Handshake AI Fellowship Jul 2026 - Present | india (Remote)

- Orchestrated Docker-based coding environments improving task reproducibility and debugging efficiency by 30 percent.
- Developed Python automation scripts to validate AI agent outputs, increasing testing accuracy and reducing manual review time.

PROJECTS

Graph Neural Networks for Financial Fraud Detection

- Developed a heterogeneous Graph Neural Network system utilizing PyTorch Geometric to detect fraudulent transactions across multi-node graphs.
- Integrated an explainability component featuring GNNExplainer and attention visualization to enhance model interpretability and identify suspicious subgraph patterns.
- Achieved superior model performance by tackling class imbalance through advanced loss functions, resulting in a 0.94 AUC-ROC metric.

Industrial Real-Time IoT Data Streaming & Predictive Intelligence Platform

- Constructed a high-throughput Industrial IoT data ingestion pipeline leveraging Apache Kafka with Avro schema validation to ensure reliable message processing.
- Orchestrated a real-time feature engineering workflow using Spark Structured Streaming with windowed aggregations persisted in an ACID-compliant Delta Lake feature store.
- Delivered a low-latency FastAPI inference service with integrated monitoring and alerting frameworks, supporting containerized deployment and CI/CD automation.

Distributed ETL Pipeline & Data Warehouse — E-Commerce Analytics Platform

- Created a scalable distributed ETL pipeline based on Apache Spark and Medallion Architecture to process and cleanse large-scale e-commerce datasets.
- Formulated a dimensional data warehouse with star schema and Slowly Changing Dimensions to support comprehensive business analytics and customer insights.
- Implemented a full-stack analytics platform with REST API endpoints, caching, and automated testing achieving high code coverage and robust monitoring integration.

EDUCATION

